

Predoc Evaluation Task: Report

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All three tasks are implemented in Python. Each task is organized in its own folder (`task1/`, `task2/`, and `task3/`). The scripts form a reproducible pipeline, with `run_all.py` specifying the execution order. Intermediate files are stored in `work/`, final results in `output/`, and figures in `figures/`. The main computational components are tested using both observed and simulated data, including simulated examples with known parameters. The AI prompt log is provided in `prompt_log.md`.

1 Constructing a Restaurant Price Index

1.0 Exploratory Data Analysis

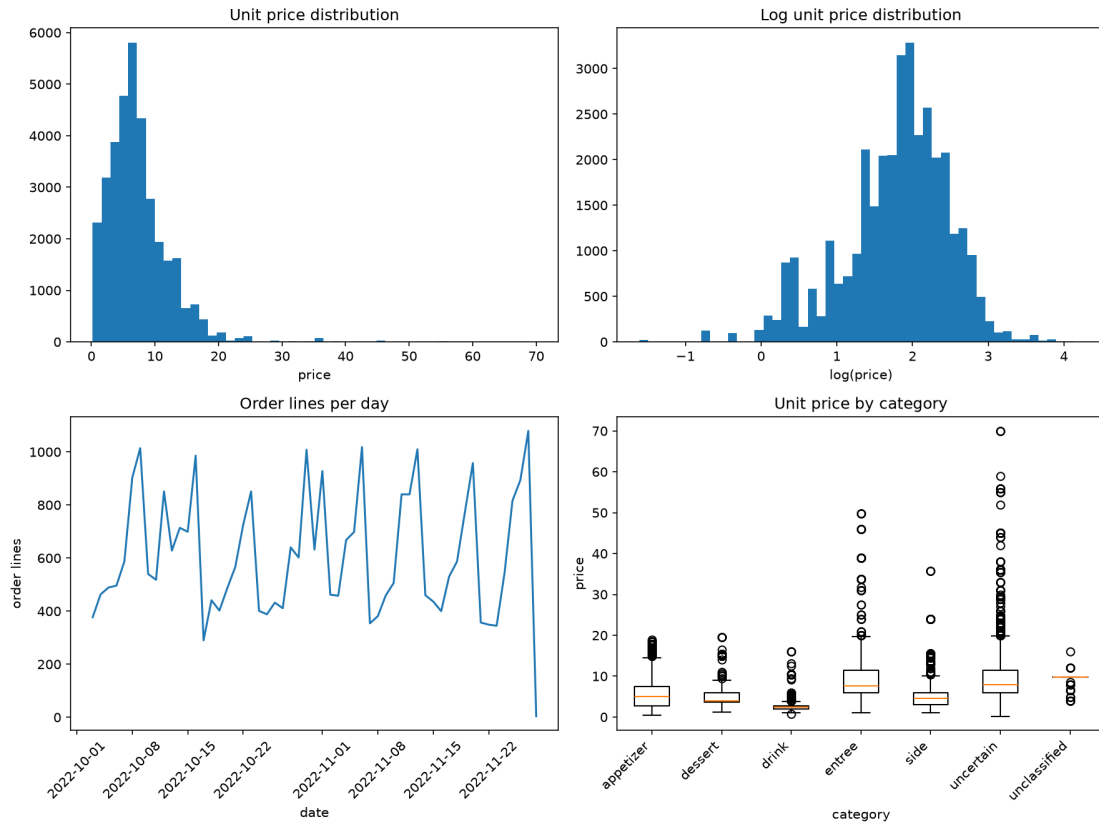


Figure 1: Exploratory analysis of restaurant prices and product categories.

Table 1: Task 1: summary statistics for restaurant order lines.

<i>Panel A: Sample structure and data quality</i>								
Raw order lines								34,752
Clean order lines								34,676
Restaurants								50
Unique dates								57
Date range							2022-10-03 to 2022-11-28	
Restaurant-product pairs								3,095
Restaurant-date pairs								2,318
Rows missing product, price, or date								76
Zero-quantity rows								52
Restaurants with no repricing								27
<i>Panel B: Variable summary statistics</i>								
Variable	<i>N</i>	Mean	SD	Min	P25	Median	P75	Max
Unit price	34,747	7.34	4.87	0.20	4.00	6.50	9.50	70.00
Quantity	34,747	1.15	0.52	0.00	1.00	1.00	1.00	10.00

Figure 1 summarizes the exploratory analysis. Unit prices are right-skewed, motivating the log transformation. Prices differ substantially across product categories, supporting the use of product fixed effects, while daily order counts vary across the sample. The data contain 34,752 order lines from 50 restaurants, and 76 rows are missing a key field. Finally, 27 restaurants do not change any product prices during the sample, so their estimated daily price levels remain constant over time.

1.1 Model and method

The goal is to decompose each order-line price into a product-specific size and a restaurant-date price index. I assume the multiplicative model

$$p_{ijt} = s_{ij} \cdot \pi_{it}, \quad (1)$$

where p_{ijt} is the unit price of product j at restaurant i on date t , s_{ij} is the size of product j relative to a typical entree at restaurant i , and π_{it} is the price of one entree-equivalent unit. Taking logs gives an exactly additive two-way fixed-effects model,

$$\log p_{ijt} = \alpha_{ij} + \delta_{it} + \varepsilon_{ijt}, \quad (2)$$

which I estimate by least squares separately for each restaurant (product dummies plus date dummies, first date as baseline, one observation per order line). Sizes are recovered as $s_{ij} = e^{\alpha_{ij}} / \bar{s}_i^e$ and indices as $\pi_{it} = e^{\delta_{it}} \cdot \bar{s}_i^e$, where $\bar{s}_i^e$ is the arithmetic mean of entree sizes at restaurant i ; this enforces the required normalization (average entree size equals 1) while leaving all fitted prices $s_{ij}\pi_{it}$ unchanged. The decomposition is only identified within a connected component of the bipartite product–date graph, so I check connectivity with a

union-find pass and normalize each component separately (anchoring to the component's entree mean where entrees exist, and to the component mean otherwise), reporting component labels in the output.

The multiplicative form is the natural choice here: relative prices of items on a menu are far more stable than their levels, restaurant-wide repricing scales the whole menu roughly proportionally, and the log-linear version turns the decomposition into a transparent, closed-form regression whose residuals directly measure deviations from the assumed structure.

1.2 Outputs

Running `python task1/run_all.py` produces the two required datasets: `product_sizes.csv`, with one row per restaurant-product, and `price_indices.csv`, with one row per restaurant-date.

1.3 Diagnostics

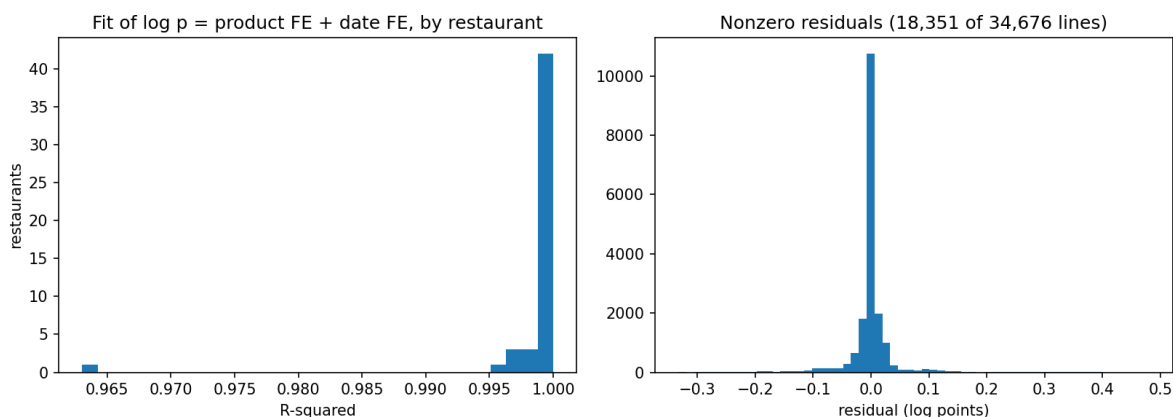


Figure 2: Task 1: within-restaurant model fit and the distribution of nonzero residuals from the product and date fixed-effects model.

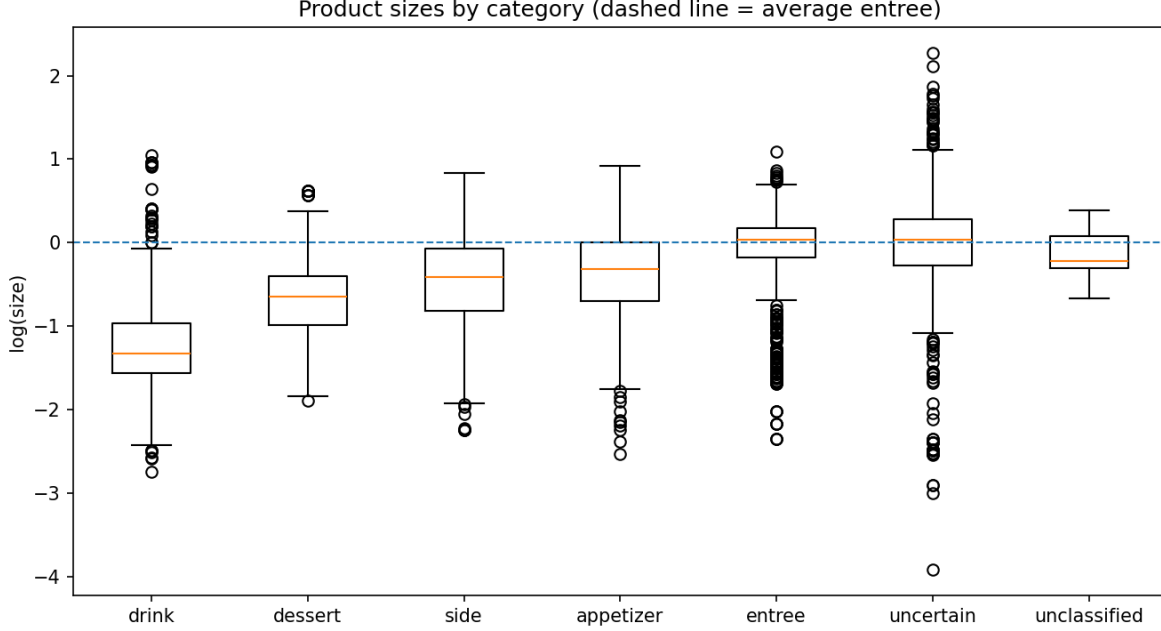


Figure 3: Task 1: distributions of estimated product sizes by category. The dashed line denotes the normalized average entree size.

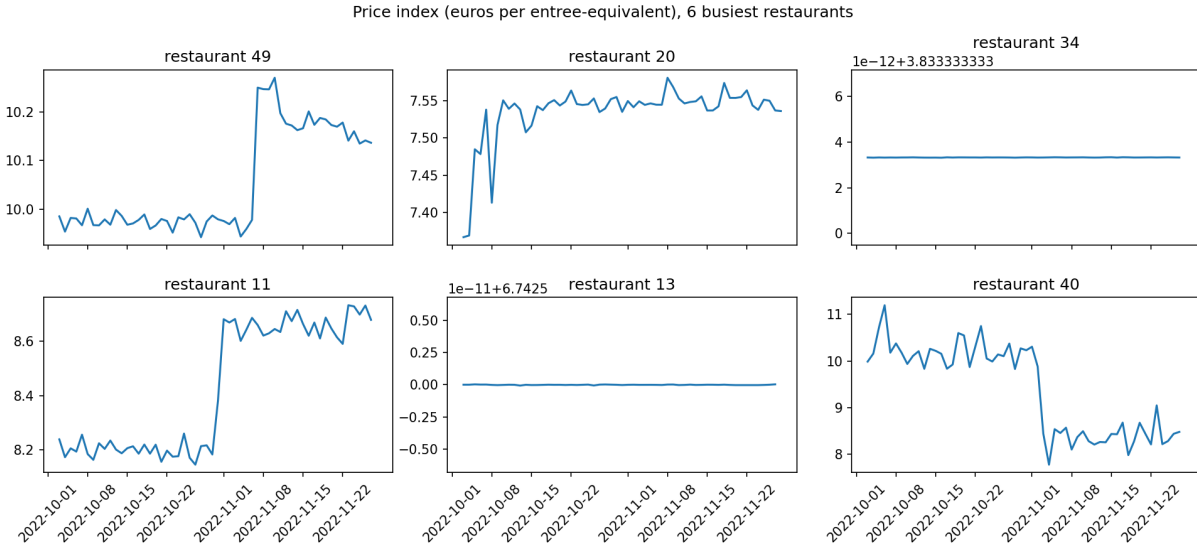


Figure 4: Task 1: estimated restaurant price indices over time for the six busiest restaurants.

I use three diagnostics to evaluate the decomposition. First, fit quality statistics check whether the estimated sizes and price indices reproduce the observed prices. The median restaurant-level R^2 is 1.000, indicating a very good fit.

Second, I compare estimated product sizes across categories. Entree sizes are centered around one by construction, while drinks, sides, and desserts are generally smaller, as expected.

Third, I plot each restaurant’s price index over time to identify unexpected jumps or unstable estimates. No restaurant is classified as highly volatile.

There are several data limitations. Twenty-seven restaurants never change prices, so their estimated price indices are flat. Three restaurants have disconnected product–date graphs and must be normalized separately by component. Three restaurants have sparse data, and fourteen have many products observed only once, making those product-size estimates less reliable.

2 Maximum Likelihood Estimation of a Mixture Model

2.0 Exploratory Data Analysis

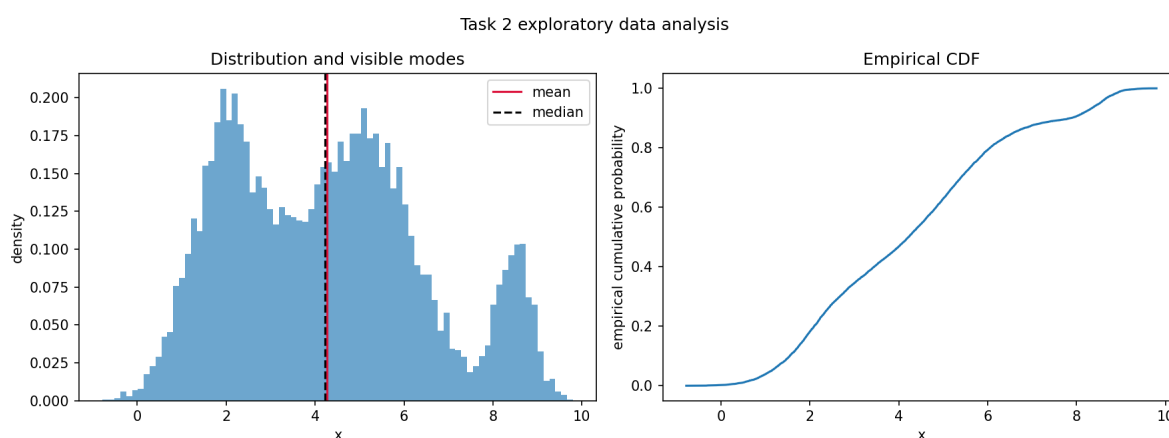


Figure 5: Task 2: empirical distribution and cumulative distribution of the mixture-model sample. The histogram reveals three visible modes.

Table 2: Summary statistics for the Task 2 mixture sample

Statistic	Value
Total observations	10,000
Non-missing observations	10,000
Missing observations	0
Mean	4.271
Standard deviation	2.220
Minimum	-0.779
1st percentile	0.475
25th percentile	2.354
Median	4.222
75th percentile	5.701
99th percentile	8.986
Maximum	9.797
Skewness	0.356
Excess kurtosis	-0.676

The empirical distribution is strongly non-normal and visibly multimodal, with three concentrations near 2, 5, and 8.5; the upper concentration is especially narrow. This shape suggests that the required two-component normal mixture may be too restrictive and that the likelihood may contain substantively different local optima. I therefore use the visual modes to construct one informed starting value and use many additional random starts to reduce sensitivity to initialization. I fit the required two-component model as the primary specification and use a three-component fit as a diagnostic comparison, evaluated using fitted densities, QQ plots, AIC, and BIC.

2.1 Log-Likelihood Function

For observations $x_1, \dots, x_n$, I compute

$$\ell(\pi, \mu_1, \sigma_1, \mu_2, \sigma_2) = \sum_{i=1}^n \log [\pi \phi(x_i; \mu_1, \sigma_1) + (1 - \pi) \phi(x_i; \mu_2, \sigma_2)].$$

The function is implemented in `task2/loglik.py` as `loglik()`.

2.2 Maximum-likelihood estimation

I maximized the log-likelihood numerically using the L-BFGS-B optimizer. Because `scipy.optimize.minimize` performs minimization, the objective supplied to the optimizer was the negative log-likelihood.

The best solution found produced the following estimates:

$$\hat{\pi} = 0.9116, \quad \hat{\mu}_1 = 3.8594, \quad \hat{\sigma}_1 = 1.8652,$$

$$\hat{\mu}_2 = 8.5166, \quad \hat{\sigma}_2 = 0.3578.$$

The corresponding maximized log-likelihood was

$$\ell(\hat{\theta}) = -21,521.395.$$

Several optimization challenges arose:

1. *Local optima.* The mixture likelihood is non-convex, so different starting values can lead to different solutions. I used 25 starting values with a fixed random seed and retained the converged result with the largest log-likelihood. Only 2 of the 25 starts reached the best solution, while 21 converged to an inferior local optimum with log-likelihood approximately $-21,602.36$.
2. *Degenerate solutions.* A normal-mixture likelihood can become unbounded when a component concentrates on one observation and its standard deviation approaches zero. I therefore imposed the lower bound

$$\sigma_k \geq 0.01 \text{ sd}(x).$$

This bound did not bind at the reported solution.

3. *Parameter constraints.* I optimized over $\log(\sigma_1)$ and $\log(\sigma_2)$ to ensure positive standard deviations, and used a logit transformation for π to ensure that $0 < \pi < 1$.
4. *Label switching.* Exchanging the two component labels leaves the likelihood unchanged. I therefore sorted the fitted components by their estimated means, so that $\hat{\mu}_1 < \hat{\mu}_2$.

Although the multi-start procedure reduces sensitivity to initial values, it cannot prove that the reported solution is the global maximum. I therefore report it as the best solution found across the 25 starts.

2.3 Model diagnostics

The empirical distribution has three visible modes near 2, 5, and 8.5, whereas the required model contains only two normal components. I assess the fitted model using density and QQ plots. I also fit a three-component model as a diagnostic comparison.

Figure 6 shows that the $K = 2$ model captures the narrow upper mode near 8.5 but uses one broad component to represent both lower modes near 2 and 5. It therefore fails to reproduce the valley between the two lower modes. In contrast, the $K = 3$ model assigns one component to each visible concentration and follows the empirical density more closely.

The QQ plots in Figure 7 give the same conclusion. The $K = 2$ QQ plot displays systematic curvature and has a quantile RMSE of 0.251 from the 45-degree line. The $K = 3$ points lie much closer to the reference line, with a quantile RMSE of only 0.012. Because both values are computed from the same observations and on the same scale, the smaller value indicates a closer distributional fit.

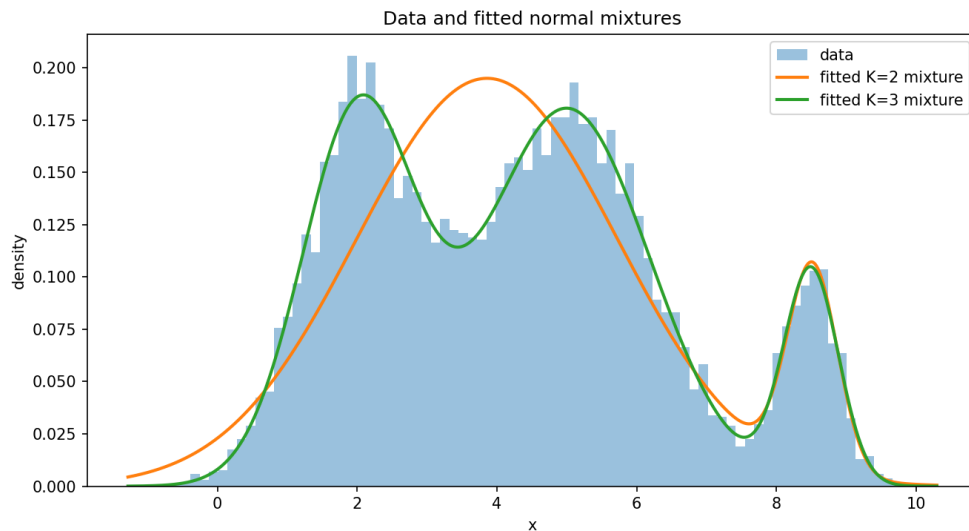


Figure 6: Observed distribution with fitted $K = 2$ and $K = 3$ normal mixture densities.

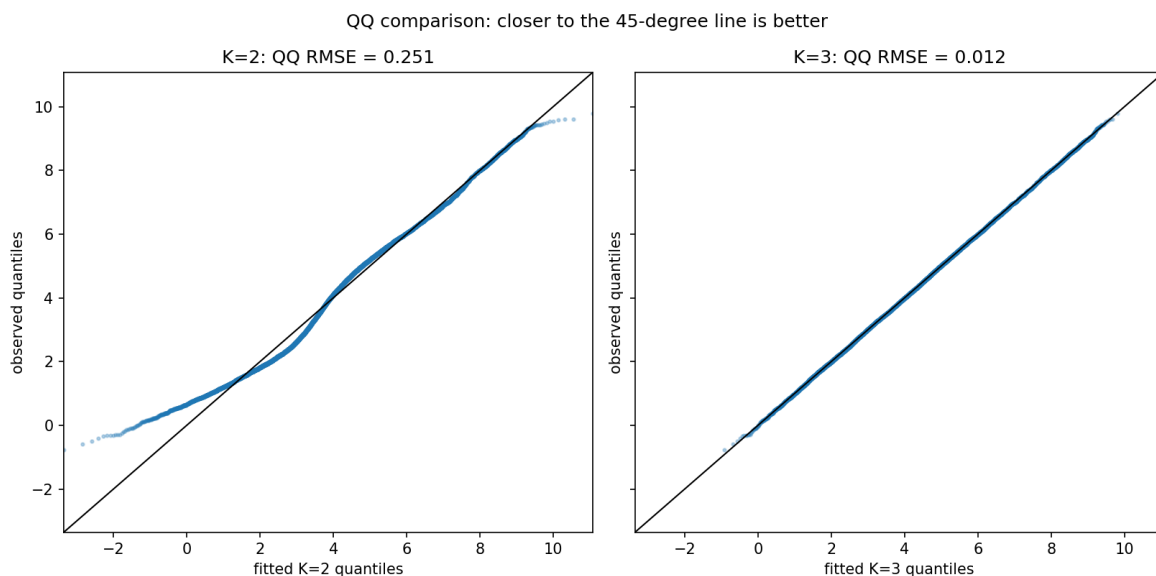


Figure 7: QQ plots for the fitted $K = 2$ and $K = 3$ mixtures. A closer match to the 45-degree line indicates better fit.

The fitted $K = 2$ model reproduces the first two sample moments closely. The sample and fitted means are both approximately 4.271, while the sample and fitted standard deviations are both approximately 2.220. However, matching the mean and standard deviation does not imply that the complete distribution is adequately fitted.

The information criteria provide an additional comparison:

Model	Log-likelihood	AIC	BIC
$K = 2$	-21,521.40	43,052.79	43,088.84
$K = 3$	-20,963.82	41,943.63	42,001.31

Both AIC and BIC are substantially lower for $K = 3$, even after penalizing its three additional parameters. Overall, the diagnostic results show that the required $K = 2$ model cannot fully capture the three visible modes in the data. I therefore report the $K = 2$ estimates as the main result required by the assignment, while using the $K = 3$ model only to illustrate the limitations of the two-component specification.

3 Do Readers' Clicks Affect Story Length?

3.0 Exploratory Data Analysis

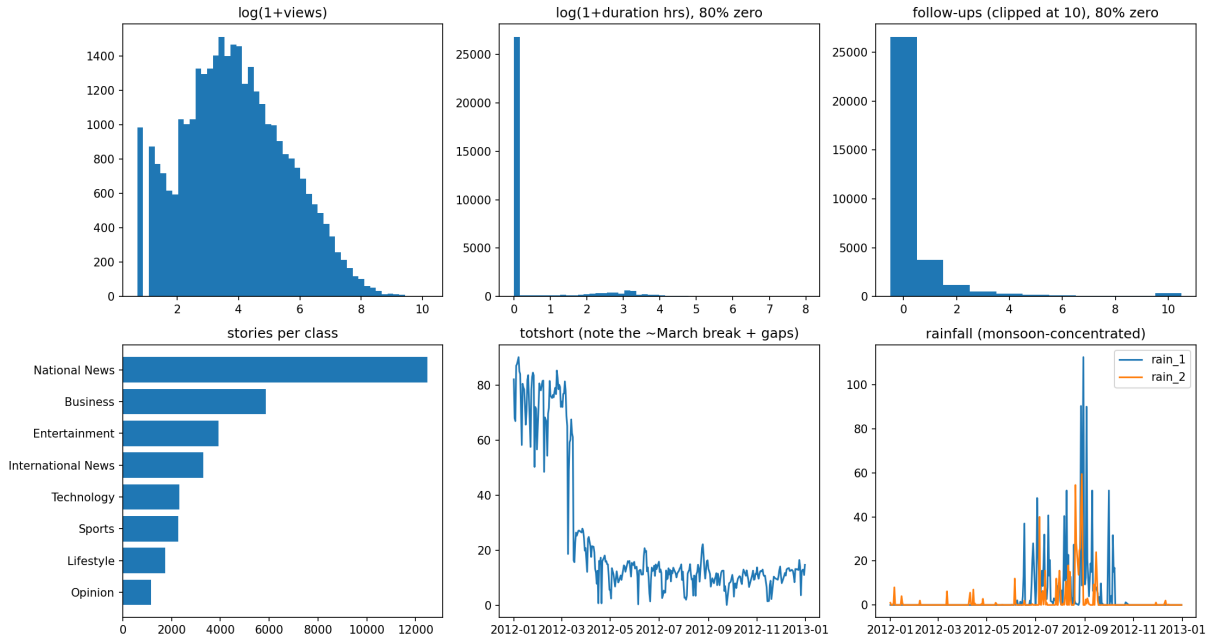


Figure 8: Task 3: exploratory overview of log views, zero-valued outcomes, story classes, and the daily instruments.

Table 3: Task 3 summary statistics for the merged analysis sample.

Variable	N	Mean	SD	Min	Median	Max	Zero share
Views	33,094	213.15	611.30	1.00	44.00	26611.00	0.0%
Duration (hours)	33,094	4.90	30.11	0.00	0.00	2904.00	80.3%
Follow-up articles	33,094	0.71	6.43	0.00	0.00	377.00	80.3%
Word count	33,094	383.20	256.62	6.00	331.00	4930.00	0.0%
Electricity shortage	33,094	24.81	25.98	0.08	12.62	90.10	0.0%
Rainfall, city 1	33,094	4.56	13.55	0.00	0.00	112.60	68.3%
Rainfall, city 2	33,094	1.77	6.92	0.00	0.00	59.50	80.3%

Figure 8 shows these distributions and also presents story class composition and the time patterns of the three instruments. Table 3 reports summary statistics for the merged sample. Duration and follow-up articles both have a median of zero and a zero share of about 80%. Their maxima are also far above their means, which shows that the positive observations have long right tails. These features motivate the $\log(1 + y)$ transformations. Views are strictly positive but also highly skewed, so I use $\log(\text{views})$.

3.1 Data Cleaning, Merge, and Variable Construction

I convert the three date variables to a common datetime format because the files use different date strings. I then merge rainfall and electricity shortage onto each story using its start date. The daily files have unique dates, so the merge is many-to-one. Rainfall covers every story date. The power data are missing for 41 days. I drop the 4,233 stories starting on those days rather than treating missing electricity shortage as zero. The final sample contains 33,094 stories.

I construct duration as the difference between the recorded end and start timestamps. It measures the editorial coverage window. I define follow-up articles as `narticles`—1 because the total includes the first article. I also construct month and day-of-week variables, log one plus word count, and a binary image indicator for use as controls.

As shown in the EDA, both outcomes contain many zeros and have long right tails. I therefore use $\log(1 + y)$ for the outcomes. Views are strictly positive but highly skewed, so I use $\log(\text{views})$. I retain large observations because the data checks do not identify them as impossible values. Finally, I verify that no story ends before it starts, that views are positive, and that every story contains at least one article.

3.2 Identification Strategy

Reader clicks may be endogenous because story importance can affect both readership and editorial investment. For example, an important story may receive more views and may also be covered for longer or receive more follow-up articles. Reverse causality is also possible if continued coverage generates additional attention to the original article. OLS may therefore not identify the causal effect of clicks.

I use three instruments for $\log(\text{views})$: daily electricity shortage and rainfall in two major cities. All instruments are measured on the story’s start date. These conditions may change

readers’ access to or attention to online news, but they are not determined by the quality of an individual story. In the first stage, I regress $\log(\text{views})$ on the three instruments, story controls, and fixed effects. In the second stage, I use the instrumented variation in views to estimate its effect separately on $\log(1 + \text{duration})$ and $\log(1 + \text{follow-ups})$.

The specification controls for first-article characteristics and includes story-class, month, and day-of-week fixed effects. The time fixed effects are important because electricity shortage and rainfall have strong trends and seasonal patterns. Standard errors are clustered by story start date because the instruments vary at the daily level.

The IV strategy requires relevance and the exclusion restriction. Relevance requires the three instruments to predict views, which I assess using a joint first-stage test. The exclusion restriction requires rainfall and electricity shortage to affect editorial investment only through views. This assumption is plausible because the instruments are external to an individual story, but it is not guaranteed. Rainfall may itself generate news, and electricity shortages may directly disrupt the newsroom. I therefore report these threats and use Anderson–Rubin inference, which remains valid when the instruments are weak.

3.3 Results and Validity

Table 4: IV estimates of the effect of log views.

	$\log(1 + \text{duration})$	$\log(1 + \text{follow-ups})$
2SLS	−0.300 (0.496)	−0.026 (0.185)
AR p -value at $\beta = 0$	0.84	0.66
AR 95% confidence set	unbounded	unbounded
First-stage statistic	0.59	

Table 4 presents the main results. The 2SLS coefficient is −0.300 for duration, with a standard error of 0.496 and a p -value of 0.545. The coefficient is −0.026 for follow-up articles, with a standard error of 0.185 and a p -value of 0.890. A 10% increase in views therefore corresponds to point estimates of approximately a 3% decrease in $1 + \text{duration}$ and a 0.3% decrease in $1 + \text{follow-ups}$. However, neither estimate is statistically significant. The standard errors are large, so the negative signs should not be interpreted as reliable causal effects.

The main concern is the weak first stage. The joint first-stage statistic is only 0.59. This means that rainfall and electricity shortage provide very little usable variation in views after the controls and fixed effects are included. The Anderson–Rubin test does not reject a zero effect for either outcome. The AR p -values are 0.84 for duration and 0.66 for follow-ups, and both 95% confidence sets are unbounded. The exclusion restriction is also uncertain because rainfall may generate news and electricity shortages may directly disrupt newsroom production. I therefore cannot determine whether clicks have a positive, negative, or zero causal effect on either outcome.

A better design would use more precise instruments. One option is hourly rainfall matched to the exact publication time. Another is an unexpected hourly internet or electricity outage in readers’ locations that does not affect the newsroom. Hourly rainfall is easier to obtain

because historical weather data are publicly available. Reader-side outage data would likely provide a more credible instrument, but they require detailed location and network records.

As a robustness check for the exclusion restriction, I could restrict the sample to story categories that are unlikely to cover weather or electricity shortages. For example, I could exclude National News and re-estimate the model using the remaining categories. This would reduce the possibility that rainfall or shortages affect follow-up production because they become news themselves. However, it would not rule out a direct effect of electricity shortages on the newsroom, and it would not solve the weak first stage. I would therefore treat this restriction as a robustness check rather than as a complete solution.